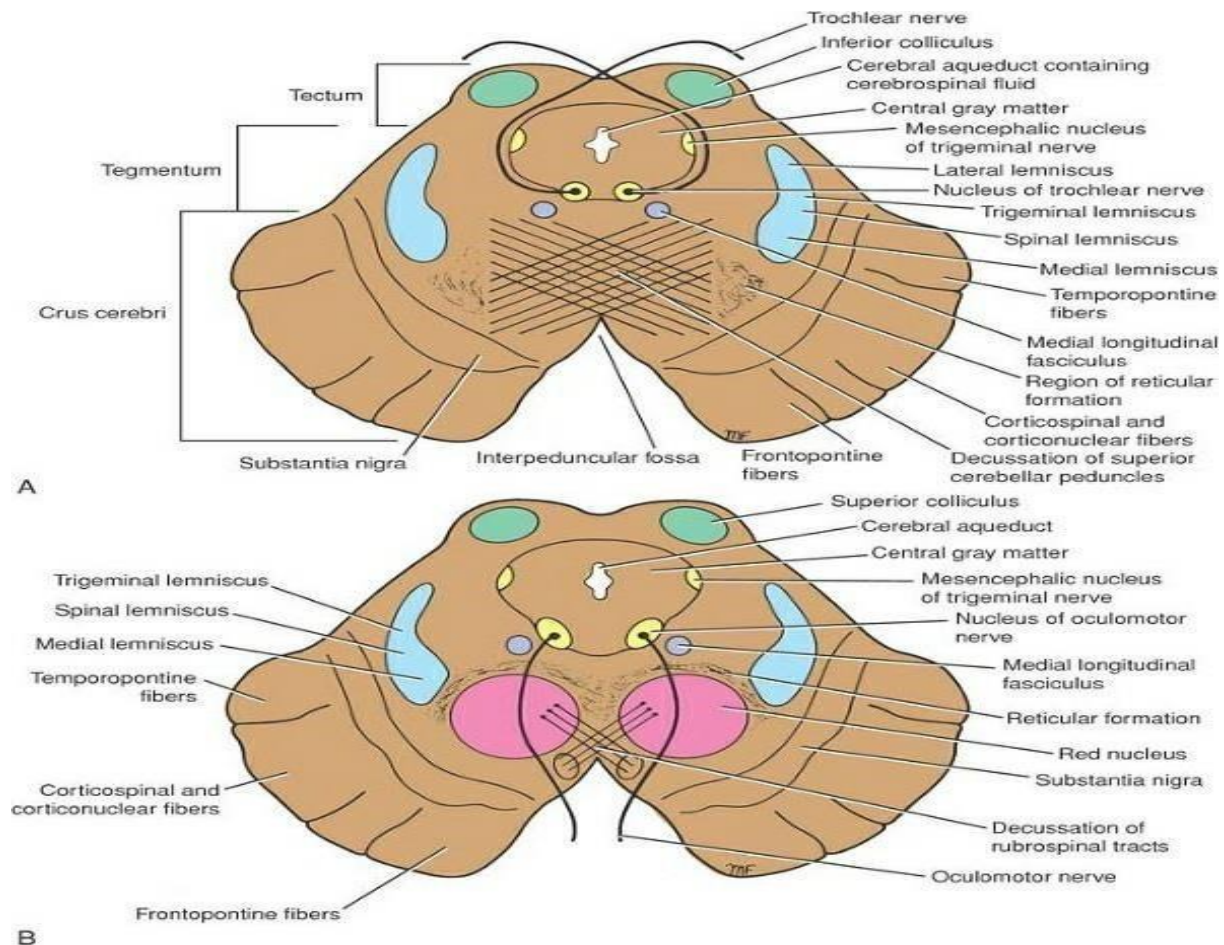




Transverse sections of midbrain at level of inferior colliculus (A) & superior colliculus (B)

Lecture (14): cerebrum I

- The cerebrum is the largest part of the brain, situated in the cranial cavity.
- It is divided into two parts: **diencephalon** and **cerebral hemispheres**.
- The **two cerebral hemispheres** are separated by a deep midline **longitudinal fissure** which is occupied by **falx cerebri** and **two anterior cerebral arteries**.
- The two cerebral hemispheres are connected by the **corpus callosum** across the midline.
- The cerebral hemispheres are separated horizontally from the cerebellum by the **tentorium cerebelli**.
- The surface of each cerebral hemisphere is thrown into a complicated series of tortuous folds called **gyri**, which are separated from each other by **sulci** or **fissures**.
- **Lobes of cerebrum**: the cerebral hemisphere is divided into **4 lobes**:
 - **Frontal lobe** lies in front of the central sulcus and above the lateral sulcus.

- **Parietal lobe** lies behind the central sulcus and above the lateral sulcus.
- **Temporal lobe** lies below the lateral sulcus.
- **Occipital lobe** lies below and behind the parieto-occipital sulcus.

- **Poles of cerebrum:**

- **Frontal pole:** the anterior end of frontal lobe.
- **Temporal pole:** the anterior end of temporal lobe.
- **Occipital pole:** the posterior end of occipital lobe.

- **The four main Sulci:**

1- Central sulcus:

- The central sulcus indents the **supero-medial border** of the hemisphere, about **1 cm behind the midpoint**.
- It runs downward and forward **across the lateral surface** of the hemisphere and ends just above the **posterior ramus of the lateral sulcus**.
- It separates the **precentral gyrus** of the frontal lobe anteriorly from the **postcentral gyrus** of the parietal lobe posteriorly.

2- Lateral sulcus:

- It lies on the **inferior and lateral surfaces** of the cerebral hemisphere.
- It consists of a **short stem** that divides **into 3 rami**.
- The **stem** arises on the **inferior surface**, and on reaching the lateral surface, it is divided into **anterior horizontal ramus, anterior ascending ramus and posterior ramus**.
- A buried part of the cerebral cortex, called the **insula** lies at the **bottom** of the lateral sulcus.
- It lodges the **middle cerebral artery**.

3- Parieto-occipital sulcus:

- It begins on the superior medial margin of the hemisphere about **2 inches anterior to the occipital pole**.
- It passes downward and anteriorly on the medial surface to meet the calcarine sulcus.
- It separates the **parietal and occipital lobes**.

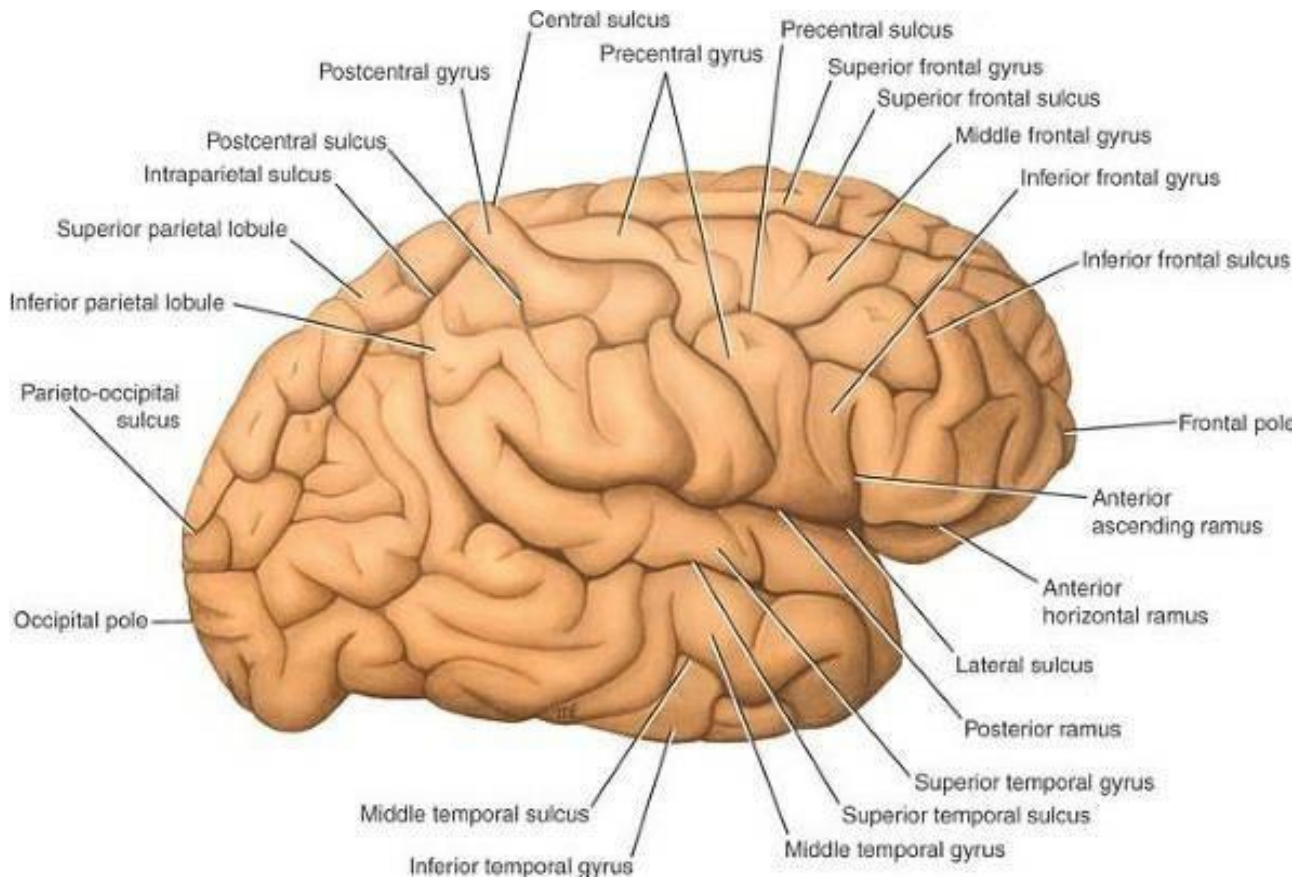
4- Calcarine sulcus:

- It is found on the **medial surface of the cerebral hemisphere**.

- It begins under the splenium of the corpus callosum, then arches upward and backward to reach the occipital pole.
- The calcarine sulcus is joined at an acute angle by the parieto-occipital sulcus.
- It lodges **posterior cerebral artery**.

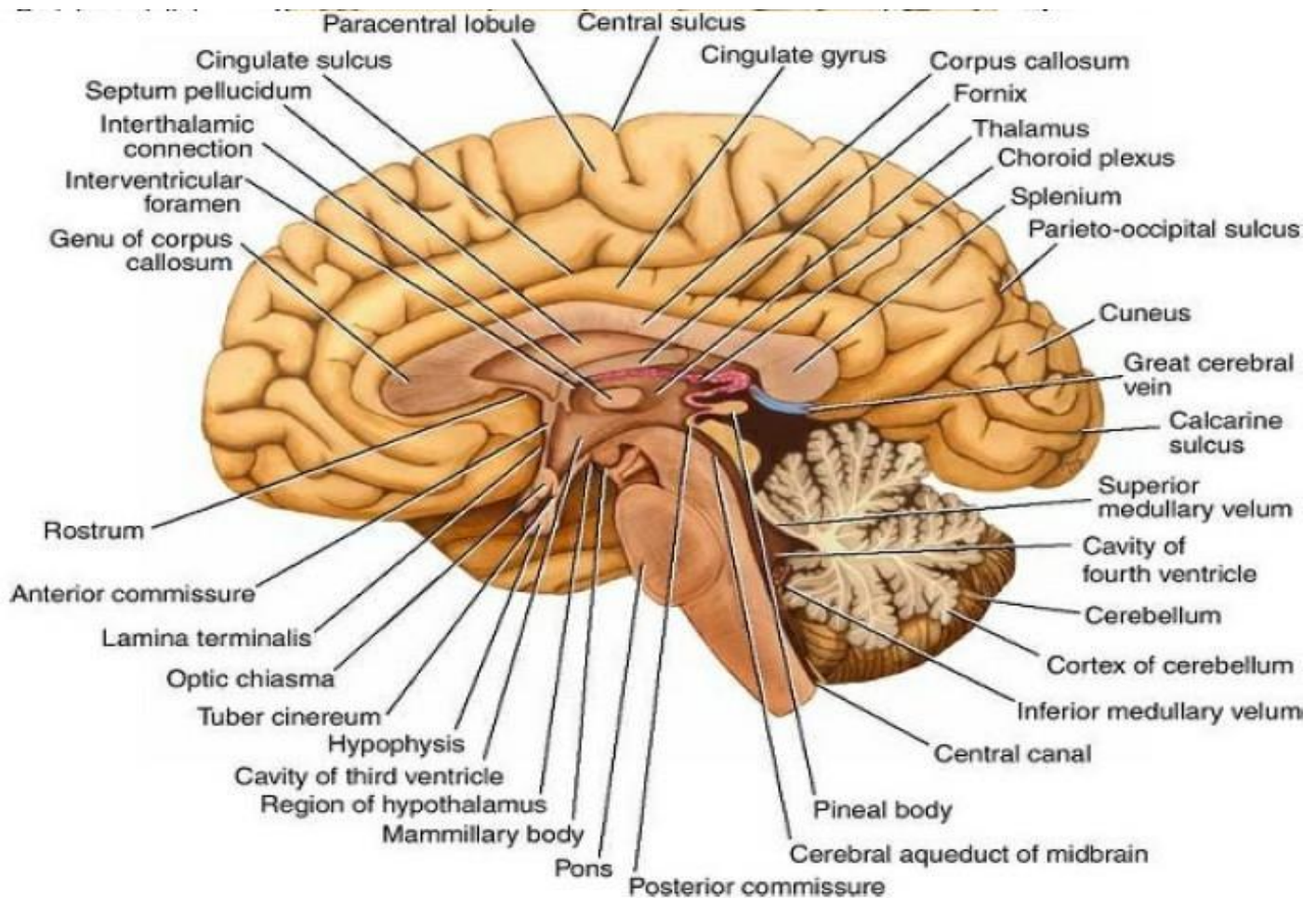
Surfaces of cerebral hemispheres:

o **Superolateral surface of the frontal lobe** has the following features:



(Superolateral surface)

Medial surface:



- The **corpus callosum**, which is the largest commissure of the brain, forms a striking feature on this surface.
- The **cingulate gyrus** lies above the corpus callosum. The gyrus is separated from the corpus callosum by the callosal sulcus.
- **Cingulate gyrus** is separated from the superior frontal gyrus by the **cingulate sulcus**.
- The **paracentral lobule** is the area of the cerebral cortex that surrounds the indentation produced by the central sulcus on the superior border. The anterior part of this lobule is a continuation of the precentral gyrus. The posterior part of the lobule is a continuation of the postcentral gyrus.
- The **precuneus** is an area of cortex bounded anteriorly by the upturned posterior end of the cingulate sulcus and posteriorly by the parieto-occipital sulcus.
- The **cuneus** is a triangular bounded above parieto-occipital sulcus, inferiorly by

calcarine sulcus.

Inferior surface:

- The stem of lateral sulcus divides this surface into the **orbital surface** of the frontal lobe and the **tentorial inferior surface** of the temporal & occipital lobes.

- On the **orbital surface**:

1) **Olfactory bulbs and tract** overlie the olfactory sulcus.

2) **Gyrus rectus** lies medial to the olfactory sulcus.

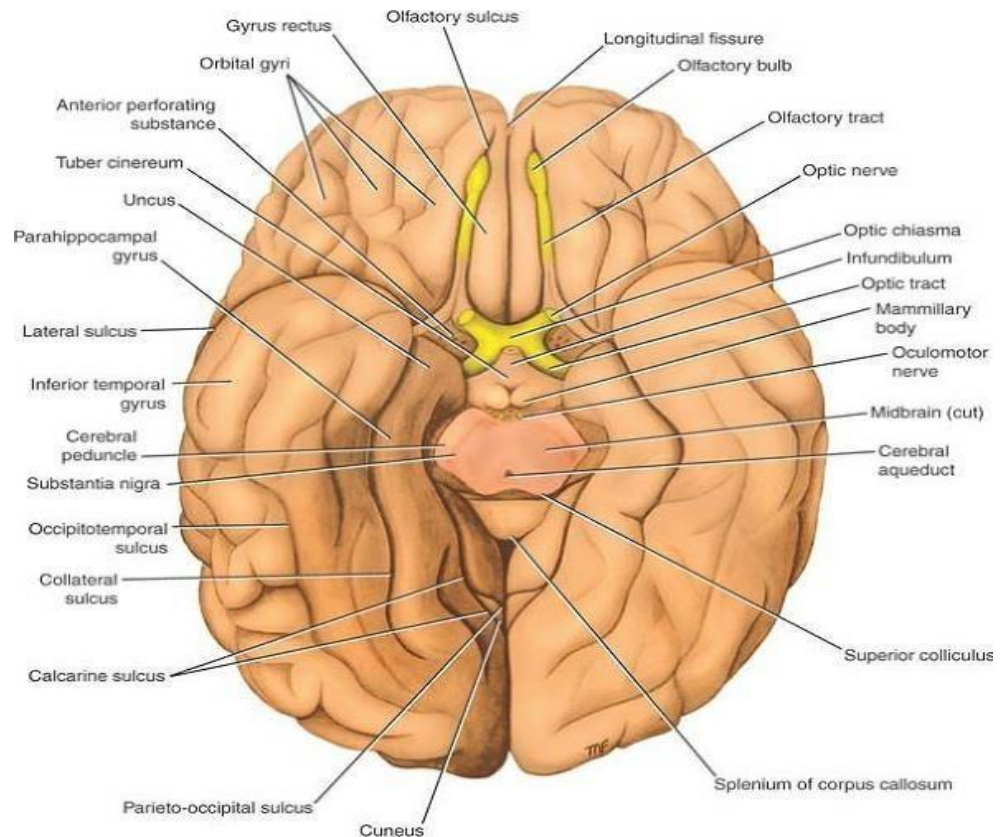
3) **Four orbital gyri** lay around H-shaped **orbital sulcus**. ○ on the **tentorial surface**:

1- **Collateral sulcus** runs anteriorly below calcarine sulcus.

2- **Lingual gyrus** lies between the collateral sulcus and the calcarine sulcus.

3- **Parahippocampal gyrus** lies anterior to the lingual gyrus. Its anterior end is called **uncus**.

4- **Occipitotemporal gyrus** is bounded medially by collateral sulcus and laterally by occipitotemporal sulcus.



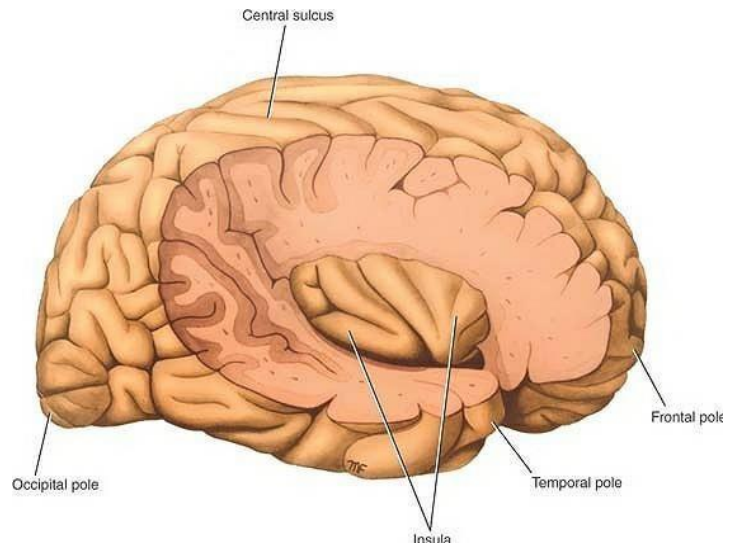
Insula:

- It is a **triangular** buried area of the cerebral cortex that lies in the bottom of the lateral sulcus.
- The insula is hidden or covered by parts of frontal, parietal and temporal lobes, called the **opercula**.
- It is composed of various long and short gyri.
 - Its **apex (limen insulae)**: is directed downwards and forwards.

- Its **base**: is surrounded by circular **sulcus**.
- The **middle cerebral artery & deep middle cerebral vein** run on the outer surface of insula. **Insula** is related deeply to extreme & external capsules and claustrum.
- It is associated with **autonomic and emotional responses** to sensory stimuli through its connections with limbic and taste areas. It is important in **pain sensation**.

Internal Structure of the Cerebral Hemisphere

- 1- **Cerebral cortex** forms a complete covering of the cerebral hemisphere. It is composed of gray matter. The surface area of the cortex has been increased by throwing it into **gyri** which are separated by **sulci**.
- 2- Masses of gray matter are in the interior of the cerebral hemisphere called the **basal ganglia**.
- 3- The myelinated nerve fibers constitute the inner **white matter**.



Cortical areas of the frontal lobe

■ **Primary motor area (Area 4):**

- **Site:** in precentral gyrus & anterior part of paracentral lobule.
- The body is represented in this area upside down as a **motor homunculus**.
- Importance: Execution of fine skilled movements of the contralateral side of the body.
- **Arterial supply:**
 - upper one inch by anterior cerebral artery.
 - The rest by middle cerebral artery.
- **Input:** thalamus, premotor area, area 3,1.2.
- **Output:** about (40%) of fibers of pyramidal tract arise from Giant Pyramidal Cells of Betz area 4.
- **Lesion:** contralateral hemiplegia (UMNL).

■ Premotor area (Area 6), secondary motor area:

- Site: anterior part of precentral gyrus in front of area 4.
- Input: from sensory cortex, thalamus, and basal ganglia.
- It projects to area 4 & cerebellum.
- Its main function is planning and storage of programs of motor activities.

■ Supplementary motor area (secondary motor area):

- Site: in the medial frontal gyrus on the medial surface, anterior to the paracentral lobule.
- Input: from thalamus & basal ganglia.
- Output: to primary motor cortex.
- Function: planning & sequencing of movements.

■ Frontal eye field (Area 8):

- Site: in the posterior part middle frontal gyrus immediately in front of premotor cortex.
- Function: control conjugated movements and voluntary scanning movements of the eye.
- It is connected to visual area of the occipital cortex & superior colliculus.
- Lesion: results in conjugate **deviation of the eyes toward the side of lesion**.

■ Motor speech area of Broca (Area 44,45):

- It is in the posterior part of inferior frontal gyrus of dominant hemisphere.
- It coordinates actions of muscles used in speech as muscles of the larynx, mouth, tongue, soft palate. Lesion: **motor (expressive) aphasia**.

■ Prefrontal association cortex (Areas 9, 10, 11 & 12):

- It includes the greater parts of the superior, middle, and inferior frontal gyri, orbital gyri & medial frontal gyrus.
- It is concerned with the makeup of the individual's personality, judgement and behavior.
- Lesion: results in frontal lobe syndrome.

Cortical areas of the parietal lobe

■ Primary somatosensory area (Area 3,1,2):

- Site: in anterior part of postcentral gyrus & posterior part of paracentral lobule.
- The body is represented in this area upside down.
- Function: perception of general somatic sensation from the contralateral side of

the body.

- Input: PLVNT & PMVNT of thalamus. Output: area 4 & secondary sensory area.
- Lesion: contralateral hemianesthesia.

▣ **Somatosensory Association area (Area 5, 7):** Site: Superior parietal lobule.

- Function: Interpretation and integration of different sensory modalities.
- Lesion: inability to recognize the felt objects (astereognosis)

🕒 **Gustatory area, Area 43:**

- At the lower end of the postcentral gyrus in the superior wall of the lateral sulcus.
- Taste sensation.

▣ **Supramarginal and Angular gyri (Areas 39, 40):** inferior parietal lobule. Interpretation of sensory data. Understanding the written language. Lesion: agraphia (inability to write) & alexia (inability to read).

Cortical areas of the temporal lobe

▣ **primary auditory area (Brodmann areas 41 and 42):**

- It lies in the middle of superior temporal gyrus.
- Conscious perception of sound. Lesion: partial deafness.

▣ **Secondary auditory area (auditory association cortex):**

- It is situated posterior to the primary auditory area in the superior temporal gyrus.
- Interpretation of sounds.

▣ **Primary olfactory cortex, (area 28):**

- **Uncus (area 28)** is center of smell lies in the anterior end of Para hippocampal gyrus.

▣ **Sensory speech area of Wernicke (Area 22):**

- Is localized in superior temporal gyrus of dominant hemisphere.
- Understanding of written & spoken language. Lesion: sensory (receptive) aphasia.

Cortical areas of the occipital lobe

1- **Primary visual area (Brodmann area 17):**

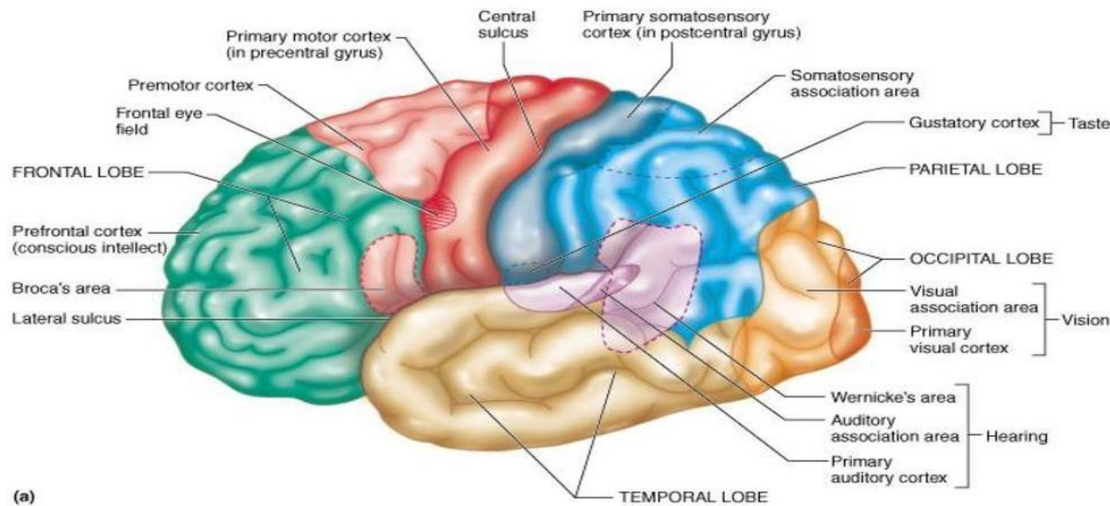
- It is situated on both lips of the calcarine sulcus & extends to the occipital pole.
- The visual cortex receives afferent fibers from lateral geniculate body (from

the ipsilateral temporal retinal fibers & contralateral nasal retinal fibers).

- Perception of vision. Lesion: Contralateral homonymous hemianopia with macular sparing.

2- Secondary visual area (Brodmann areas 18 and 19):

- Surrounds the 1ry visual area on the medial and lateral surfaces of the occipital cortex.
- Receives afferent fibers from area 17.
- Function: color vision & recognition of human faces and appreciate what you are seeing.
- Lesion: visual agnosia (inability to recognize what you see).



White matter of the cerebral hemispheres

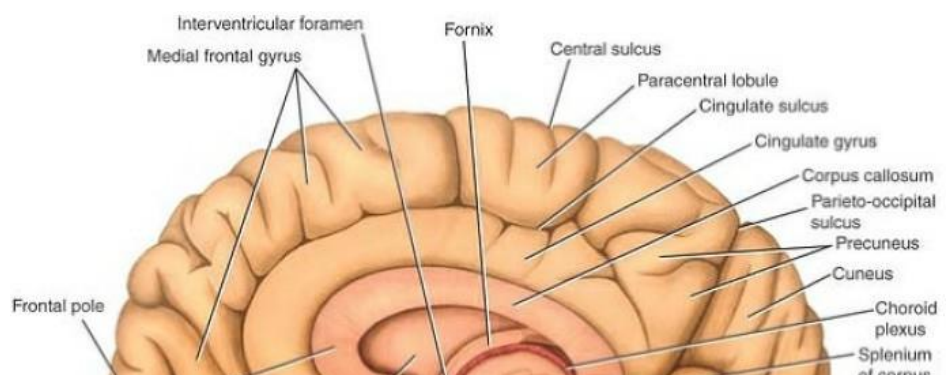
- The white matter is composed of myelinated nerve fibers.
- It can be classified into three groups according to their connections:
(1) commissural fibers. (2) association fibers. (3) projection fibers.

Commissural fibers

Definition: Commissural fibers essentially connect the corresponding regions of the two cerebral hemispheres as the following:

1- corpus callosum:

- The corpus callosum, the largest commissure of the brain, connects the two cerebral hemispheres.
- It lies at the bottom of the longitudinal fissure.
- Its upper surface is

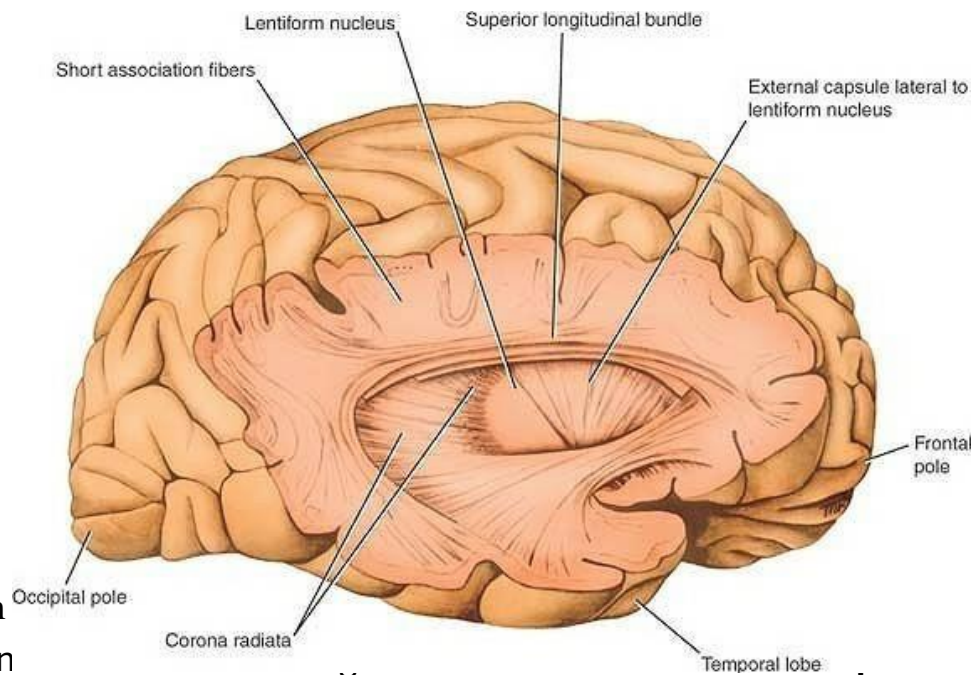


related to anterior cerebral artery & falx cerebri.

- Its lower surface is related to septum pellucidum & lateral ventricle.

- Parts:

- **Rostrum:** is the thin anterior end part, which
- **Genu:** is the curved an **minor** which connects frontal lobes.
- **Body:** arches laterally forming radiation of corpus callosum. Some posterior fibers forming roof & lateral wall of posterior horn of lateral ventricle, called **tapetum** which connects temporal lobes.
- **Splenium:** thickened posterior portion. Fibers of splenium arch backward forming **forceps major** that connects occipital lobes.



2- **Anterior commissure:** Crosses the midline in the lamina terminalis. Connects **olfactory bulbs & temporal lobes**.

3- **Posterior commissure:** It connects **two superior colliculi, pretectal & Edinger Westphal nuclei**.

4- **Fornix:**

- Connects two hippocampal gyri. The fornix constitutes efferent nerve fibers of the hippocampus.
- **Parts:** The nerve fibers collect to form the **fimbria**, which arches above the thalamus to form **two posterior columns of the fornix**. The two columns which connect with each other by hippocampal commissure to form **the body of the fornix**.
- **Two anterior columns** are downward continuation of the body that terminate on the mammillary bodies of hypothalamus.

Association fibers

- They are nerve fibers that essentially connect various cortical regions within the same hemisphere.

- They are divided into:

1- **Short association fibers**

lie beneath the cortex and connect adjacent gyri.

2- **Long association fibers**

connect distant cortical areas:

- **Uncinate fasciculus** is formed of an arched fiber, connecting frontal lobe with temporal pole.
- **Cingulum** lies within the cingulate gyrus. It connects the frontal and parietal lobes with parahippocampus of temporal lobe.
- **Superior longitudinal fasciculus** connects the frontal, parietal, occipital and temporal lobes.
- **Inferior longitudinal fasciculus** runs connect the occipital lobe to the temporal lobe.

Projection fibers

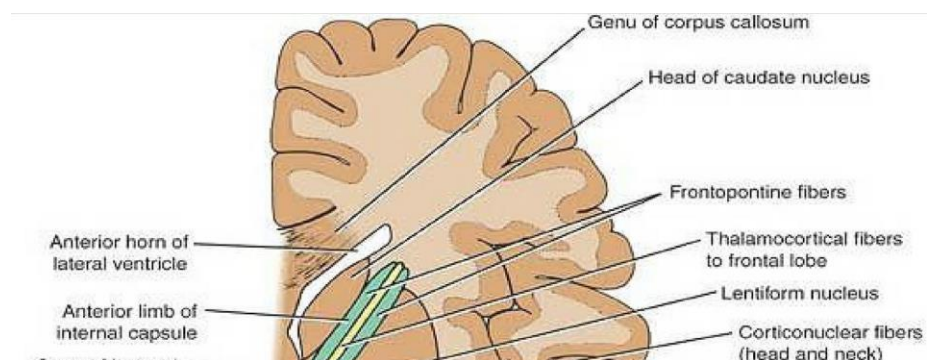
- Afferent and efferent nerve fibers passing to and from the entire cerebral cortex to subcortical centers.
- These fibers form the **internal capsule**.

Internal capsule:

- **Definition:** It is a compact band of fibers passing into or out of the cerebral cortex. It is a V-shaped.
- **Continuity:** it is continuous above with the corona radiata and below with the crus cerebri of midbrain.
- **Parts & Relations** has 3

parts:

- 1- **Anterior limb:** between the head of caudate medially & lentiform nucleus laterally.
- 2- **Genu:** the angle of the



internal capsule between caudate nucleus, lentiform nucleus & thalamus.

3- **Posterior limb:** is divided into:

- **Lenticulothalamic** part: between thalamus, medially & lentiform nucleus, laterally.
- **Retrolenticular** part: behind lentiform nucleus.
- **Sublenticular** part: below lentiform nucleus.

▣ **Fibers forming the internal capsule:**

1- **Anterior limb:**

- **Descending fibers:** frontopontine fibers.
- **Ascending fibers:** anterior thalamic radiation from the anterior and medial thalamic nuclei.

2- **Genu:**

- **Descending fibers:** corticonuclear tract which pass from the cerebral cortex to the motor nuclei of cranial nerves in the brainstem.

3- **Lenticulothalamic part:**

- **Descending fibers:** corticonuclear fibres and corticospinal tract in the anterior $\frac{2}{3}$.
- **Ascending fibers:** Superior thalamic radiation in the posterior half.

4- **Retrolenticular part:**

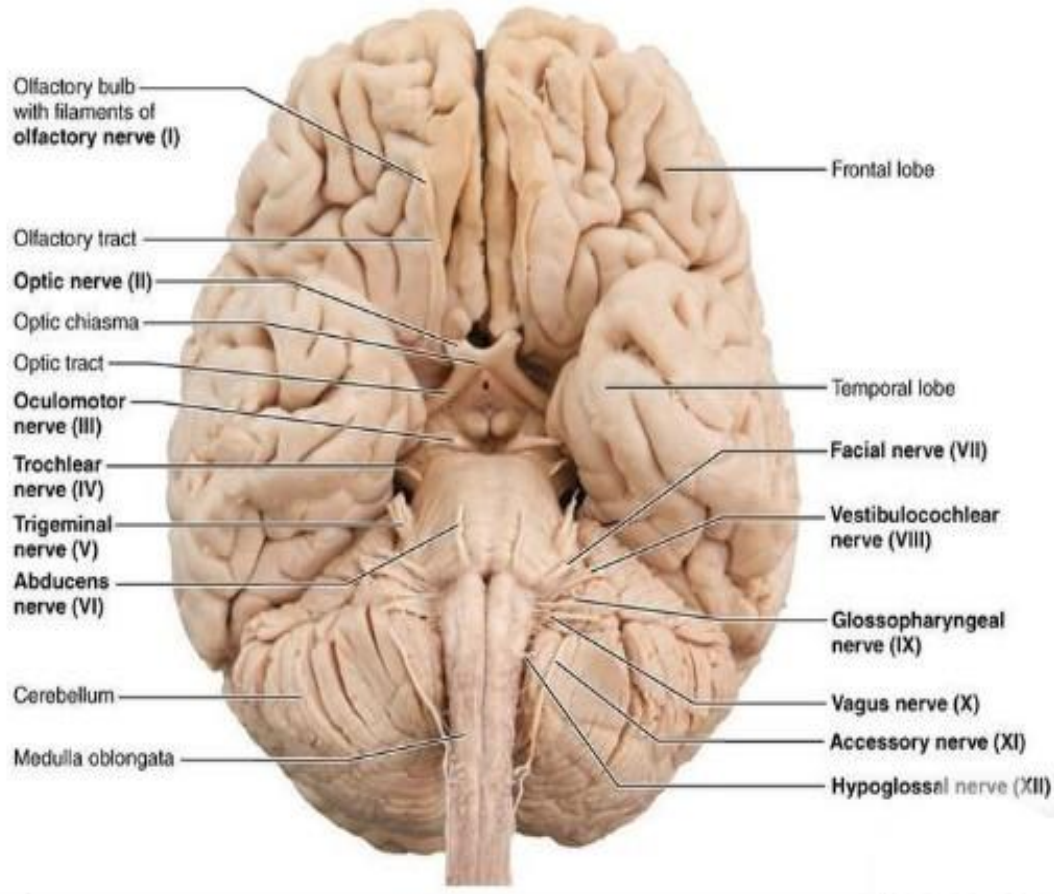
- **Descending fibers:** occipitopontine fibers.
- **Ascending fibers:** optic radiation from LGB.

5- **Sublenticular part:**

- **Descending:** temporo-pontine fibers.
- **Ascending:** auditory radiation from MGB.

▣ **Blood supply:** from the following branches of internal carotid artery.

- 1- **Anterior cerebral artery** supplies anterior part of anterior limb.
- 2- **Middle cerebral artery:** Supplies posterior part of anterior limb, genu & lenticulothalamic part.
- 3- **Anterior choroidal artery:** Supplies sublenticular, retro lenticular parts and ventral half of posterior limb.



Lecture (15): cerebrum II

Basal Nuclei (Basal Ganglia)

- The **basal ganglia (BG)** consist of **subcortical interconnected nuclei** which are deeply situated inside the white matter of the cerebral hemispheres.
- They are involved in the **initiation** and
- **Components of BG:**
 - **Caudate nucleus, putamen, globus pallidus, amygdaloid nucleus & nucleus accumbens** (in the telencephalon).

